INORGANIC CUMULATIVE EXAM

March 27, 1999

Attached are seven Tables and Figures (or parts thereof).

Choose five (5) of these and for each one you choose, write an explanation of the main idea(s) presented by the figure/table. Your answers will be graded with respect to the depth and breadth of your explanations.

1

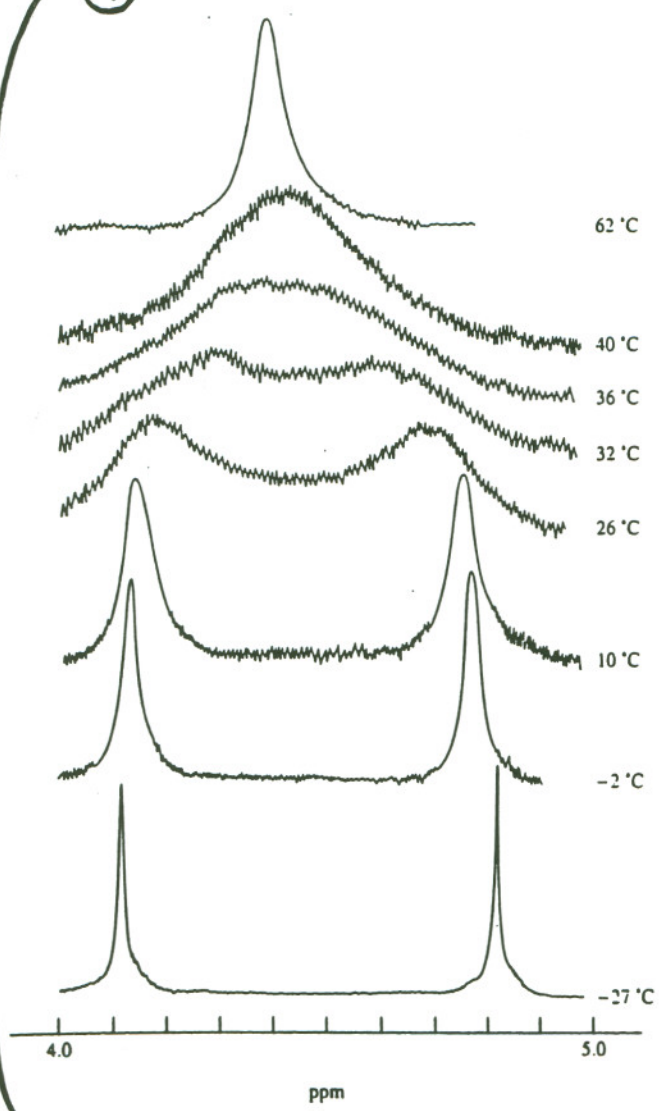


Fig. 15.52 Proton NMR spectra of $(\eta^1\text{-}C_5H_5)_2(\eta^5\text{-}C_5H_5)_2Ti$ from −27 to +62 °C. The separate

2

Table 13.4

Rate constants for the reaction of $[Co(NH_3)_5(H_2O)]^{3+}$ with X^{n-} in water at 45 °C[a]

X^{n-}	k (M^{-1} s^{-1})
NCS^-	1.3×10^{-6}
$H_2PO_4^-$	2.0×10^{-6}
Cl^-	2.1×10^{-6}
NO_3^-	2.3×10^{-6}
SO_4^{2-}	1.5×10^{-5}

[a] Basolo, F.; Pearson, R. G. *Mechanisms of Inorganic Reactions*, 2nd ed.; Wiley: New York, 1967.

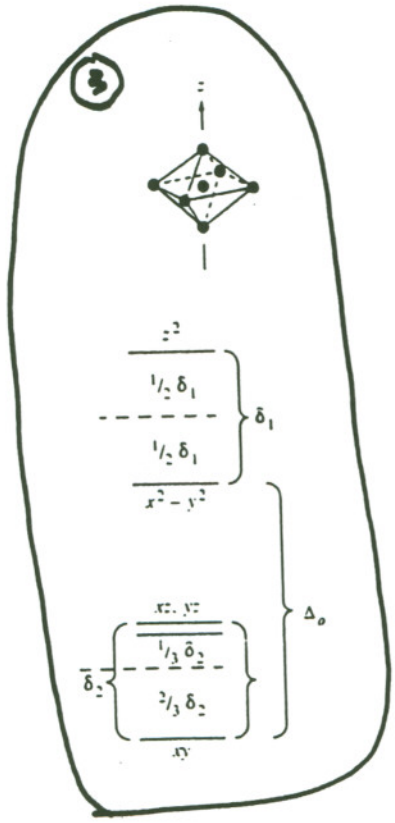
Table 13.5

Rate constants for the reaction of $[Co(NH_3)_5X]^{m+}$ with H_2O[a]

X^{n-}	k (s^{-1})
NCS^-	5.0×10^{-10}
$H_2PO_4^-$	2.6×10^{-7}
Cl^-	1.7×10^{-6}
NO_3^-	2.7×10^{-5}
SO_4^{2-}	1.2×10^{-6}

[a] Basolo, F.; Pearson, R. G. *Mechanisms of Inorganic Reactions*. 2nd ed.; Wiley: New York, 1967.

3



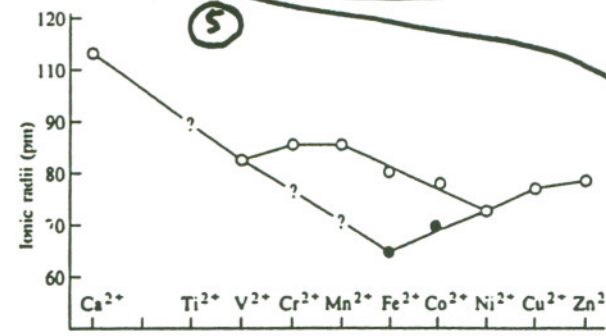
4

Table 11.14

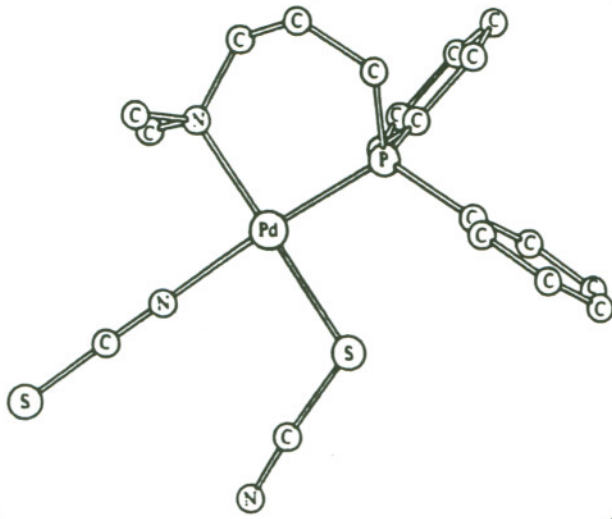
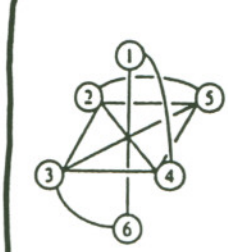
Infrared carbonyl stretching frequencies (cm^{-1}) for some $W(CO)_5L$ complexes[a]

L	$A_1^{(1)}$
Me_2NCHO	1847
Me_2CO	1847
Et_2O	1908
H_2S	1916
Ph_2S	1930
Me_2S	1932
$CyNH_2$[d]	1894
pyridine	1895
aniline	1916
CH_3CN	1931
H_3P	1921
Me_3P	1947
Me_2PhP	1947
$MePh_2P$	1947
Ph_3P	1942
$(MeS)_3P$	1946
$(EtO)_3P$	1959
$(PhO)_3P$	1965
I_3P	1981
Br_3P	1991
Cl_3P	1990
F_3P	2007

5



6



7